AI Large Model Image Usage Instructions

This tutorial mainly introduces the differences between the motherboard AI large model image and the car's factory image.

Important Notes

This chapter requires the use of our pre-configured large model image, which can be obtained from the download section AI_pure_systemfile: AI_pure_Nano_4G_20250915.zip

1. The default image in the TF card received with the car package is the car's factory image, only suitable for running the AI large model development chapter. To run the examples in this chapter, please burn the AI_pure_Nano_4G_20250915.img image file.

2. Because Ollam requires a lot of memory to run large models, motherboards with limited memory can only use models with small parameters, or even fail to run them. Sometimes, when memory usage is at its limit, the performance will be poor. For a better experience of this feature, please refer to the online large model chapter. **
3. The commands in the AI Large Model chapter are mostly run within Docker containers. Before running a command, you need to enter a Docker container. When running a command within a Docker container, there may be some lag, which is normal. **
System Information

Based on Jetson Nano Jetpack 4.6.3 Ubuntu 18.04

Username: Jetson

Password: Yahboom

System Environment

- Ollama
- Docker
- Open WebUI
- Offline large model storage path: /usr/share/ollama/.ollama/models

```
Due to SD (TF) card limitations, we did not download all models from the tutorial
to the AI_pure_Nano_4G_20250915.zip image system. Some high-parameter versions of
models in the tutorial have been deleted. Users can delete and download the
models themselves.
```

Parameter Description

Jetson Nano (4GB RAM): Runs models with 3B or fewer parameters

While the above conclusions are not entirely accurate, they can be used as a reference!

```
Ollam's official website states that the following specifications apply to
running models: 7B parameter models require at least 8GB of available RAM; 13B
parameter models require at least 16GB of available RAM; and 33B parameter models
require at least 32GB of available RAM.
```